

E.3 / PAPER 1 / QUESTION 39

25 April 2024 / TZ2 (file label) / HL Paper 1 / original Q39 / 2224-6707

- 39.** A radioactive substance has a half-life of 5 hours and a decay constant λ . At time $t = 20$ hours N nuclei are present in the sample.

What was the activity of the sample at time $t = 5$ hours?

- A. $4N\lambda$
- B. $8N\lambda$
- C. $\frac{4N}{\lambda}$
- D. $\frac{8N}{\lambda}$